AGRITECH & RURAL BIO-ECONOMY PLATFORM

SETUMOOLYA (सेतुमूल्य)

"Every Root Deserves Its Worth"

Comprehensive System Architecture & Complete Module Specifications

End-to-End Backend & Frontend Implementation Blueprint

Direct Value Transfer for Himalayan & High-Value Wild Herbs

SetuMoolya is an institutional-grade digital platform designed to dismantle exploitative 5-hop middleman chains in high-value wild herbs and botanicals (Kutki, Gucchi, Ashwagandha, Tulsi). By connecting rural foragers and Farmer Producer Organisations (FPOs) directly with global verified B2B buyers through **photo/lab-grade quality certification**, **smart escrow contracts**, and **sustainable wild-crop quota governance**, SetuMoolya returns up to 5x–10x more economic value to rural indigenous households while safeguarding fragile biodiversity.

Stack: Supabase (PostgreSQL + Auth + RLS), Node.js/Express, Next.js / React **Edition:** Hackathon & Production Roadmap
Architecture: Web-First Microservices / Modular Monolith **Version:** 1.0.0 | October 2026

1. Executive Summary & Market Problem

1.1 The Critical Market Failure in High-Value Botanicals

India's alpine and forest belts produce some of the world's most sought-after botanicals, wild mushrooms, and medicinal herbs. However, the current value chain is severely broken across three critical dimensions:

1. Asymmetric Bio-Compound Knowledge: Rural foragers harvest high-value herbs without knowing their active chemical compound concentrations (e.g., *Picrosides I & II* in Kutki, *Curcumin* in Haldi, or *Withanolides* in Ashwagandha). Middlemen exploit this information void by buying harvests at generic bulk rates and selling them to global pharmaceutical/cosmetic houses at premium compound-potency rates.

2. The 5-Hop Exploitative Middleman Chain: The prevailing distribution channel hops through: *Forager* → *Local Village Trader* → *Regional Distributor* → *Exporter/Wholesaler* → *Global Retail Brand*. Each hop consumes a significant margin while contributing zero post-harvest value. The original harvester retains less than 5% to 15% of the end retail worth.

3. Ecological Degradation & Wild Over-Harvesting: Cash-starved foragers harvest prematurely to generate immediate liquidity. This strips the wild rootstock before seed dispersal occurs, endangering wild species like *Picrorhiza kurroa* (Kutki) and wild Morel mushrooms (Gucchi).

1.2 Documented Price Disparity Analysis

Real-world field data reveals astonishing markups between farm-gate realization and branded retail value:

Herb / Crop	Active Bio-Marker	Forager Farm-Gate	SetuMoolya Fair Gate	Global / Branded Retail	Value Delta
Gucchi (Wild Himalayan Morel)	Ergosterol & Polysaccharide complex	₹5,000 – ₹10,000 / kg	₹20,000 – ₹28,000 / kg	₹45,000 – ₹65,000 / kg	6x – 10x Markup
Kutki (Picrorhiza kurroa)	Picroside I & II (>8%)	₹1,350 – ₹2,300 / kg	₹5,500 – ₹7,500 / kg	₹25,000 – ₹35,000 / kg	12x – 20x Markup
Ashwagandha Root	Withanolides (>5%)	₹300 – ₹500 / kg	₹1,200 – ₹1,800 / kg	₹20,000+ / kg (Capsules)	40x – 60x Markup
Tulsi Leaf (Holy Basil)	Eugenol & Ursolic Acid	₹80 – ₹150 / kg	₹350 – ₹550 / kg	₹1,500 – ₹2,500 / kg (Tea)	10x – 20x Markup
Wild Curcumin Turmeric	Curcuminoid Concentration (>5%)	₹100 – ₹160 / kg	₹400 – ₹750 / kg	₹15,000+ / kg (Extract)	100x Markup

1.3 The SetuMoolya Solution: "One System, Four Layers"

Layer 1: Farmer & FPO Producer System

Voice-first, vernacular crop listing, real-time minimum fair prices based on bio-potency, seasonal quotas, and direct UPI/bank payouts.

Layer 2: Verified B2B Buyer Sourcing Portal

Global and domestic brands browse standardized, lab-certified lots, request physical samples, and view verified chemical Certificates of Analysis (COA).

Layer 3: Direct B2B Smart Escrow Marketplace

Legally binding digital sales contracts, escrow deposit locks, milestone disbursement upon delivery, and automated multi-split payouts directly to farmers.

Layer 4: Wild-Crop Protection & Provenance Engine

Regional harvest quotas, seasonal moratorium calendars, root-sparing audits, and QR-code farm-to-shelf consumer transparency.

2. User Roles & RBAC Security Matrix

SetuMoolya enforces strict Role-Based Access Control (RBAC) backed by Supabase Auth and PostgreSQL Row Level Security (RLS) policies. Every persona operates within specific, tamper-resistant permission boundaries:

Role Key	Persona Description	Primary Objectives	Permissions & Data Scope
farmer	**Forager / Herb Cultivator** Indigenous village collectors	Log harvests, view fair pricing, monitor harvest quotas, receive direct payments	Create harvest entries; view own quota status; view personal bank payout passbook; view crop fair price benchmarks.
fpo	**Farmer Producer Organisation** Village cooperative management	Aggregate forager batches into bulk lots, dispatch lab samples, assign payouts	Aggregate multiple farmer entries into commercial lots; create B2B listings; dispatch samples to labs; manage local warehouse storage.
buyer	**Verified B2B Buyer** Global pharma, nutraceutical & wellness brands	Browse verified lots, request samples, place bulk orders, fund escrow	Browse verified catalog; view lab COAs; order sample boxes; execute B2B contracts; fund escrow balances; track freight.
lab	**Testing Laboratory** Certified AYUSH / NABL test facilities	Test bio-compounds (HPLC/GC-MS), issue tamper-evident COAs	Receive sample aliquots; record moisture, active bio-markers, heavy metal levels; issue digitally signed Certificates of Analysis (COAs).
logistics	**Cold-Chain & Freight Partner** Domestic transport & customs clearance	Update waybills, verify phytosanitary docs, provide tracking telemetry	Update shipment checkpoints; verify temperature/humidity cold-chain logs; upload customs and phytosanitary clearance documents.
admin	**System SuperAdmin / Agronomist** Platform governance & conservation NGO	Set regional harvest quotas, approve KYC, mediate disputes, platform health	Full system access; calibrate bio-compound fair pricing formulas; set geo-zone quota limits; mediate escrow disputes; platform fee settings.

2.1 Authentication Architecture in Supabase

User credentials and identities are securely managed via Supabase Auth (OAuth / Email+Password / Phone OTP). Upon registration, a database trigger automatically synchronizes user identity with `public.profiles`, maintaining role classifications, KYC verification status, organization details, and encrypted banking information.

Security Implementation Highlights:

- **JWT Verification Middleware:** All protected backend Express routes authenticate incoming Bearer tokens using `supabase.auth.getUser(token)`.
- **Fine-Grained Role Guards:** Declarative middleware `authorize('buyer')` or `authorize(['farmer', 'fpo'])` blocks unauthorized privilege escalation.
- **PostgreSQL RLS Policies:** Even direct database queries are restricted at the database engine level based on `auth.uid() = id`.

3. Complete Backend Modules Specification

The SetuMoolya backend is designed as a modular, scalable Node.js/Express system coupled to Supabase PostgreSQL, Storage, and Realtime services. Below is the detailed architectural breakdown of all 10 backend modules:

MODULE BE-01: Auth, Identity & Role-Based Access Control (RBAC)

COMPLETED & VERIFIED

Primary Function: Secure user registration, authentication, token refresh, and role enforcement across all six platform personas.

Key Database Tables: `auth.users`, `public.profiles`, `kyc_documents`.

Key API Endpoints:

- `POST` `/api/auth/register`
- `POST` `/api/auth/login`
- `POST` `/api/auth/refresh`
- `GET` `/api/auth/me`
- `PUT` `/api/auth/me`

Core Business Logic:

- Supabase Auth session creation with custom user metadata.
- Automated PostgreSQL trigger syncs profile record upon signup.
- Admin-only KYC approval endpoint updating verification status.

MODULE BE-02: Crop Catalog & Dynamic Fair-Pricing Calculator

CORE VALUATION ENGINE

Primary Function: Maintains botanical taxonomies, active bio-compound benchmarks, and calculates real-time fair farm-gate pricing.

Key Database Tables: `crops`, `bio_marker_standards`, `market_price_benchmarks`.

Key API Endpoints:

- `GET` `/api/crops`
- `GET` `/api/crops/:id/pricing-calculator`
- `POST` `/api/crops/standards (Admin)`

Algorithmic Pricing Formula:

```
FairPrice = BaseGradePrice * (ActualBioMarker% / BaselineBioMarker%) * (1 - (ActualMoisture% - IdealMoisture%) * MoisturePenaltyFactor)
```

MODULE BE-03: Micro-Harvest Intake & FPO Lot Aggregation Engine

SUPPLY AGGREGATION

Primary Function: Records individual forager collections and batches them into standardized commercial lots with unique batch IDs and QR tracking.

Key Database Tables: `harvest_entries`, `fpo_lots`, `lot_harvest_mapping`.

Key API Endpoints:

- `POST` `/api/harvests/intake`
- `POST` `/api/lots/aggregate`
- `GET` `/api/lots/:lotId/qr-manifest`

Core Features:

- Captures gross weight, tare weight, moisture sensor readings, and harvest GPS.
- Generates cryptographic Lot UUID and generates printable warehouse QR codes.
- Logs individual farmer contribution percentages for escrow revenue split.

MODULE BE-04: Digital Quality Grading & Lab Verification Engine

VERIFICATION & COA

Primary Function: Manages physical lab sample custody, HPLC/GC-MS chemical test results, and issues tamper-evident digital Certificates of Analysis (COAs).

Key Database Tables: `lab_samples`, `lab_test_reports`, `test_parameters`.

Key API Endpoints:

`POST` `/api/lab/samples/intake`

`POST` `/api/lab/reports/issue`

`GET` `/api/lab/reports/:coaId/verify`

Security & Verification:

- Generates SHA-256 hash of lab test parameters to prevent post-issue tampering.
- Classifies quality lot tier: Grade AAA (Export Pharma), Grade AA (Nutra), Grade A (Standard).
- Auto-generates official PDF COA stored in Supabase Storage buckets.

3. Complete Backend Modules Specification (Contd.)

MODULE BE-05: Direct B2B Marketplace & Smart Contract Engine

TRANSACTION ENGINE

Primary Function: Powers verified lot listings, buyer Requests for Quote (RFQs), sample requests, and digital legally binding purchase agreements.

Key Database Tables: `marketplace_listings`, `sample_orders`, `b2b_contracts`, `rfq_bids`.

Key API Endpoints:

`GET` `/api/marketplace/lots`

`POST` `/api/marketplace/sample-request`

`POST` `/api/marketplace/contracts/generate`

Core Workflows:

- Lot listings automatically link active lab COA and minimum fair price index.
- Sample box ordering with tracking to buyer R&D laboratories.
- Generates digital purchase agreement with stipulated purity thresholds.

MODULE BE-06: Milestone Escrow & Multi-Split Payment Engine

FINANCIAL SETTLEMENT

Primary Function: Guarantees secure buyer payments, holds funds in neutral escrow, and automates multi-split payouts directly to farmer and FPO bank accounts.

Key Database Tables: `escrow_accounts`, `escrow_milestones`, `payment_disbursements`.

Escrow State Machine:

1. **ESCROW_FUNDED:** Buyer deposits 100% contract value.
2. **DISPATCH_TRIGGERED:** Logistics scans batch; 20% advance released to FPO for transit costs.
3. **DELIVERY_CONFIRMED:** Buyer inspects lot at destination; remaining 80% released.
4. **AUTO_SPLIT:** Instant payout split: 88% to farmers (via UPI/NEFT), 9% to FPO, 3% platform commission.

Key API Endpoints:

`POST` `/api/escrow/deposit`

`POST` `/api/escrow/milestones/release`

`GET` `/api/escrow/ledger/:contractId`

MODULE BE-07: Wild-Crop Protection, Quotas & Conservation Engine

SUSTAINABILITY & COMPLIANCE

Primary Function: Prevents over-harvesting of endangered alpine herbs through geo-fenced harvest quotas, seasonal bans, and replanting verification.

Key Database Tables: `harvest_quotas`, `seasonal_moratoriums`, `geo_zones`.

Key API Endpoints:

`GET` `/api/conservation/quotas/status`

`POST` `/api/conservation/quotas/check`

`PUT` `/api/conservation/moratoriums (Admin)`

Conservation Circuit Breaker:

- Strict total seasonal weight limits per species per village cooperative.
- Hard lock: Marketplace automatically rejects batch intake if quota is exhausted.
- Calendar-based harvesting moratoriums during seed dispersal and flowering cycles.

MODULE BE-08: End-to-End Cryptographic Provenance & QR Engine

TRACEABILITY

Primary Function: Builds an immutable chronological audit trail from the exact Himalayan forest coordinates to the final retail shelf QR code.

Key Database Tables: `provenance_events`, `qr_records`.

Key API Endpoints:

`POST` `/api/traceability/log-event`

`GET` `/api/traceability/public/:qrCode`

Event Trail Ledger:

`HARVESTED → AGGREGATED → LAB_CERTIFIED → PACKED → EXPORTED → RETAIL_ACTIVE`

Includes cryptographic checksums to guarantee complete supply chain integrity.

4. Complete Frontend Modules Specification (Web Portals)

SetuMoolya is built web-first, fully responsive across desktop, tablet, and mobile browsers. Below is the specification of all 6 distinct frontend portals:

PORTAL FE-01: Universal Multi-Persona Auth & Onboarding Flow

ALL USERS

Target Users: Harvesters, FPO Managers, B2B Buyers, Lab Technicians, Logistics Partners.

Key Screens & UI Components:

- **Role Selection Hub:** Visual persona cards with role-specific feature previews.
- **Smart Registration Form:** Conditional fields based on selected role (e.g. Bank IFSC for farmers, GSTIN & IEC code for export buyers).
- **KYC Verification Upload:** Document drop-zone with instant image preview.

UX & Accessibility Features:

- Instant role-based redirect upon login to dedicated dashboard.
- Full support for high-contrast visibility in bright outdoor sunlight.

PORTAL FE-02: Farmer & FPO Producer Workbench

PRODUCER INTERFACE

Target Users: Village Foragers, Cooperative Supervisors, Aggregation Centers.

Key Screens & UI Components:

- **Voice & Quick-Entry Intake Pad:** Simplified, big-button harvest logger with local vernacular names (Kutki, Gucchi, Ashwagandha).
- **Live Fair-Price Ticker:** Shows today's fair benchmark price based on current market bio-compound values.
- **Harvest Quota Meter:** Circular progress ring displaying remaining sustainable seasonal harvest allowance.
- **FPO Lot Consolidation Table:** Checkbox-driven aggregation tool that merges micro-batches into 100kg/500kg commercial lots.
- **Direct Payout Passbook:** Transparent digital passbook showing escrow disbursements and pending releases.

Field Ergonomics:

Designed with large touch targets (min 48px), minimal typing requirements, audio readout options, and offline data cache fallback for remote hilly areas.

PORTAL FE-03: Global B2B Buyer Sourcing Portal

COMMERCIAL BUYERS

Target Users: Procurement directors at Ayurveda, Nutraceutical, Pharmaceutical & Gourmet food brands.

Key Screens & UI Components:

- **Bio-Compound Filter Marketplace:** Sliders allowing buyers to filter lots by chemical purity (e.g. *Picrosides* ≥ *7.5%*, *Curcumin* ≥ *5.0%*).
- **Interactive Lot Details View:** High-res photo inspection, moisture charts, harvest village bio, and live COA download.
- **1-Click Sample Request Modal:** Dispatches 50g-100g sealed test samples to corporate R&D facilities.
- **Escrow Checkout Terminal:** Deposit contract balance into secure escrow with automated multi-milestone release schedule.
- **Shipment Tracking Radar:** Real-time cold-chain temperature telemetry and freight progression.

Enterprise Capabilities:

Multi-currency pricing display (INR, USD, EUR), automated commercial invoice and Certificate of Origin generator, export-ready documentation package.

PORTAL FE-04: Quality Testing Laboratory Workbench

LAB TECHNICIANS

Target Users: Certified AYUSH / NABL testing chemists.

Key Screens & UI Components:

- **Incoming Sample Intake Queue:** Barcode/QR scanning to check in sealed physical samples.
- **Analytical Data Entry Form:** Form fields for HPLC peak areas, active compound %, heavy metals (PPM), moisture content.
- **COA Generator & Digital Signature:** Generates tamper-proof branded Certificate of Analysis with digital signature stamp.

Assurance Controls:

Automated out-of-spec warnings if moisture exceeds safe storage thresholds (e.g. >12% for dried Kutki rhizomes).

PORTAL FE-05: Public Provenance & Storytelling Portal

CONSUMER TRANSPARENCY

Target Users: End consumers scanning QR codes on finished supplement or gourmet packages.

Key Screens & UI Components:

- **Harvester Hero Banner:** Real photo and village story of the local forager who gathered the raw herb.
- **Interactive Elevation & Forest Map:** Visualizing the high-altitude origin (e.g., 3,400m altitude in Chamoli, Uttarakhand).
- **Verified Chemical Purity Scorecard:** Consumer-friendly explanation of the batch's active compound potency.
- **Wild Harvest Sustainability Seal:** Proves that the batch adhered to sustainable harvest quotas and root-sparing practices.

Brand Impact:

Transforms commodity raw herbs into authentic, premium, high-trust consumer stories, justifying luxury retail valuations.

PORTAL FE-06: Platform Admin & Conservation Command Center

SUPERADMIN & GOVERNANCE

Target Users: SetuMoolya operations leads, Forest Department liasons, and NGO agronomists.

Key Screens & UI Components:

- **Regional Conservation Heatmap:** Real-time tracking of wild crop harvest volumes against ecological quotas.
- **Middleman Bypass Metric Dashboard:** Live calculation of additional income generated for farming families vs traditional middleman payouts.
- **Escrow & Dispute Arbiter:** Inspection logs, sample dispute resolution, and milestone release overrides.

Platform Governance:

Dynamic adjustment of platform commission rates, system health monitoring, and user KYC verification approvals.

5. Database Entity Relationship Model & System Flow

5.1 Supabase Database Entity Architecture

The system utilizes a relational PostgreSQL architecture in Supabase, leveraging Foreign Keys, JSONB metadata, and Row Level Security:

Table Name	Primary Key	Key Foreign Keys	Core Data Attributes
`profiles`	id (UUID)	auth.users.id	email, full_name, phone, role, organization_name, location (JSONB), kyc_status, bank_details (JSONB)
`crops`	id (UUID)	-	botanical_name, vernacular_name, primary_bio_marker, standard_bio_percentage, base_fair_price_inr, season_months
`harvest_quotas`	id (UUID)	crop_id	geo_zone, annual_max_kg, current_harvested_kg, season_year, is_quota_locked
`harvest_entries`	id (UUID)	farmer_id → profiles.id, crop_id → crops.id	weight_kg, moisture_percentage, harvest_date, gps_coordinates, photo_url, payout_status
`fpo_lots`	id (UUID)	fpo_id → profiles.id, crop_id → crops.id	batch_number, total_weight_kg, average_moisture, status ('aggregating', 'tested', 'listed', 'sold'), qr_code_id
`lab_reports`	id (UUID)	lot_id → fpo_lots.id, lab_id → profiles.id	bio_marker_percentage, heavy_metals_ppm, quality_grade ('AAA', 'AA', 'A'), coa_pdf_url, sha256_hash
`b2b_listings`	id (UUID)	lot_id → fpo_lots.id, fpo_id → profiles.id	asking_price_per_kg, min_order_kg, available_weight_kg, listing_status, sample_available
`b2b_contracts`	id (UUID)	listing_id, buyer_id → profiles.id, fpo_id	quantity_kg, agreed_price_per_kg, total_amount_inr, contract_status, contract_pdf_url
`escrow_transactions`	id (UUID)	contract_id → b2b_contracts.id	total_funded, advance_released, final_released, farmer_split_amount, fpo_split_amount, platform_fee
`provenance_events`	id (UUID)	lot_id → fpo_lots.id, actor_id → profiles.id	event_type, location_name, timestamp, metadata (JSONB), hash_signature

5.2 The 8-Step Harvester-to-Global Buyer Value Journey

1. **Identify Crop & Sustainable Harvesting Rules**
   Foragers access seasonal harvest guidance, root-sparing rules, and quota limits via the Farmer Portal.
2. **Micro-Intake & Primary Processing at Village Center**
   Harvest is cleaned, dried, weighed, and logged into the portal with moisture sensor data and photo grading.
3. **FPO Batch Aggregation into Commercial Lots**
   The village FPO consolidates small forager lots into standardized commercial batches (e.g. 200 kg Kutki rhizome lot).
4. **Independent Chemical Lab Testing (AYUSH / NABL)**
   Sealed sample aliquots are tested via HPLC for active compound % (e.g. 8.4% Picrosides). Official digital COA is generated.

5. **Direct B2B Marketplace Listing & Fair Price Discovery**
   Lot is published to verified global B2B buyers with lab certificate, provenance map, and transparent pricing.
6. **Sample Validation & Escrow Deposit**
   Buyer verifies 50g sample, executes smart contract, and deposits 100% of order value into neutral escrow.
7. **Tracked Logistics & Inspection Acceptance**
   Lot is dispatched in tamper-evident sealed packaging. Buyer scans QR and accepts shipment upon delivery.
8. **Automated Multi-Split Bank Transfer to Farmers**
   Escrow releases payment automatically: 88% disbursed directly to foragers' bank accounts, 9% to FPO, 3% platform commission.

6. Implementation Roadmap & Hackathon Milestones

6.1 Phase 1: Hackathon Core Prototype (Current Phase)

Milestone Objectives:

- **Auth & RBAC Foundation:** Full multi-role authentication with Supabase and Express middleware (Completed).
- **Crop Catalog & Fair Price Engine:** Static catalog with dynamic pricing calculation for Kutki, Gucchi, and Ashwagandha.
- **FPO Lot Aggregation & Listing Demo:** Mock batch aggregation, batch QR code generation, and marketplace lot listing.
- **Simulated Escrow & Direct Payout:** Visual transaction flow showing buyer escrow deposit and automated multi-split payout calculation.
- **Public QR Provenance Screen:** Mobile responsive web page demonstrating what a consumer sees upon scanning the batch QR code.

6.2 Phase 2: Pilot Deployment (Months 1–3)

Milestone Objectives:

- Partner with 1 high-altitude Himalayan FPO in Chamoli / Pithoragarh (Uttarakhand).
- Onboard 50 wild foragers harvesting Kutki and Gucchi.
- Integrate with 1 certified AYUSH testing lab for official digital COA uploads.
- Integrate SMS/WhatsApp Bot fallback for offline forager alerts and price broadcasts.
- Execute 3 pilot B2B direct sales to domestic Ayurvedic nutraceutical brands.

6.3 Phase 3: Commercial Scale & Global Export (Months 4–12)

Milestone Objectives:

- Expand to 15 FPOs across Himachal Pradesh, Uttarakhand, and Ladakh.
- Multi-currency international escrow checkout (USD / EUR / INR) with automated customs & phytosanitary documentation.
- IoT cold-chain telemetry integration with BLE temperature loggers attached to bulk export shipping containers.
- State Forest Department integration for automated wild harvesting permits and quota compliance reporting.

6.4 Summary Impact Projection

3.5x – 6x Average Net Income Uplift for Rural Harvesters	**100%** Traceable Farm-to-Shelf Provenance & Lab Purity
4 Middlemen Completely Bypassed in the Traditional Chain	**Zero** Over-Harvesting Violations via Strict Quota Engine